

Part 1

1)

Software architecture is the fundamental design of an entire system comprising the elements included in the system, their functions and how they interact with each other. It is the high level overview of the entire system and how every component within it works together.

2)

1. The development team achieves higher productivity as the development process has a proper, well destined structure and timeline.
2. Achieve higher quality software products through careful consideration of needs, perspectives as well as capabilities of all stakeholders.
3. The development process includes a lower margin of risk as risk management is seen as a core process within a standardized system.

3)

- Conflict between developers
 - The developers in charge of different sections of the project or two or more developers from the same section disagree with each other and disrupt each other.
 - Ex: A front-end developer changes the code another front-end developer wrote, because he thinks the logout button should be blue instead of red.
- Missed deadlines
 - The developers do not have a proper plan, therefore they make a quality software, but miss the deadline, making it a failed endeavor.
 - Ex: The government commissioned a software company to create a software to manage a fuel crisis in a week. The team delivers in a month, but the government has already collapsed.
- Over budget
 - The software product is created, but it is over the allowed budget.
 - Ex: A software development team is commissioned by a small startup. The startup can only afford \$100. The team delivers, but the project costed \$200.

Part 2

4)

Any individual or party/collection of individuals who have an interest in the software system, including but not limited to developers, maintainers and customers.

5)

Stakeholder	Concern
Bank	Absolute security and near-perfect user verification
Bank Customers	Perform transactions
Software developers	Produce a quality software that meets the expectations of all stakeholders or with valid compromises while being on time and not exceeding the allocated budget
Internet Service Providers	Provide consistent access without overloading total available bandwidth

Part 3

View	Description
Logical View	Focuses on system functionality and objects
Process View	Focuses on system performance and runtime behavior
Development View	Focuses on source code organization and modules
Physical View	Focuses on hardware and deployment environment

6)

classes;

1. food
2. delivery
3. restaurant
4. paymentOption

7)

1. browsing restaurants
2. making payments

8)

1. databases to store restaurant details
2. graphical front-end to browse food items
3. payment gateway

9)

1. data warehouse
2. in house server
3. cloud infrastructure ex: AWS

Part 5

1. Joe visits the food delivery system and searches a restaurant
2. The system queries the database
3. Database returns the restaurant if its available, or a 'not-available' message otherwise.
4. The system relays the message to Joe. If the restaurant is available Joe proceeds to inspect the menu. If not, Joe searches another name.

10)

Allows the developer to catch an issue or requirement that is visible in one architectural view, but not in another.